ProLancer

An AI-Enhanced Freelancer Marketplace

A Project Synopsis Report Submitted To

SAGE UNIVERSITY, INDORE

Bachelor of Technology

in

Computer Science & Technology

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PROJECT SYNOPSIS

Project Title : ProLancer

An AI-Enhanced Freelancer Marketplace for Secure Project Collaboration and Smart Talent Matching**.**

1. Problem Statement

- The AI "Slop" and Proposal Saturation Dilemma.
- The "Pay-to-Win" Barrier (paying before earning).
- The "Scope Creep & Ghosting Nightmare.

2. Description of the Problem Statement

a) **The AI "Slop" and Proposal Saturation Dilemma:** Generative AI allows low-quality applicants and automated bots to instantly flood job posts with hundreds of hyper-polished, fake proposals within minutes. This creates massive buyer fatigue for clients who cannot tell real experts apart from copy-pasting bots, causing true human talent to get buried.

b) **The "Pay-to-Win" Barrier (Paying Before Earning):** Legacy platforms aggressively monetize freelancers by charging steep 10%–20% commissions and forcing them to buy expensive digital tokens just for the chance to apply for a job. This creates a highly unfair "feast-or-famine" barrier.

c) **The "Scope Creep" & Ghosting Nightmare:** Freelancers face severe financial risk when clients demand extra free work outside the agreed contract or completely disappear after a project is submitted. Traditional marketplaces use slow, manual dispute systems that take weeks to resolve, leaving a freelancer's hard-earned money trapped helplessly in escrow.

3. Objective and Scope of the Project

Objectives

1. Enable freelancers to showcase portfolios and bid on relevant projects.
2. Provide secure contract, milestone, and payment management tools.

Scope of the Project

1. Core Marketplace (Baseline Scope)

- **User Profiles:** Freelancer profiles with portfolio, skills, and reviews.
- **Milestone Contracts:** Contract creation, milestone tracking, and work submission.
- **Escrow Payments:** A Stripe-integrated gateway to hold client funds, release payouts, and auto-generate invoices.

2. Advanced Features/Scope (Anti-Spam & Automation Scope)

- **Proof-of-Work Verification:** Live video recordings and native skill tests to completely stop AI bot proposal spam.
- **SaaS Subscription Model:** A flat monthly fee for clients instead of taking a commission cut from freelancers.
- **Self-Executing Escrow:** An AI-monitored chat to flag scope creep and an auto-release timer to bypass client ghosting.

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4. Methodology

The project follows the **Agile Software Development Methodology**.

1. Requirements

- **Backlog:** Define rules for escrow payouts, OpenAI chat tracking, and visual Skill Trees.
- **Roles:** Map access levels for Freelancers, Clients, and Admins.

2. Design

- **UI/UX:** Wireframe the dual dashboards, interactive tree graphs, and the code sandbox.
- **Database:** Structure the MongoDB collections with privacy filters for blind auditions.

3. Development

- **Core Systems:** Code the Node.js API, React components, and Socket.io real-time chat.
- **Integrations:** Connect Stripe Connect pipelines and set up secure background Docker containers and other APIs.

4. Testing

- **Quality Check:** Test the 48-hour escrow countdown timers and OpenAI text-parsing edge cases.
- **Fixes:** Debug data flow bugs across the interactive skill graph nodes.

5. Deployment

- **Live Launch:** Host your production React bundle on Render/Vercel.
- **Backend:** Deploy the Express server and Docker environment to AWS or Render.

6. Review

- **Evaluation:** Analyse platform performance logs and test-sandbox execution speeds.
- **Iteration:** Refine the AI risk prediction parameters based on real contract data.

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5. Limitations

- Third-Party API Dependency
- High Server Infrastructure Costs

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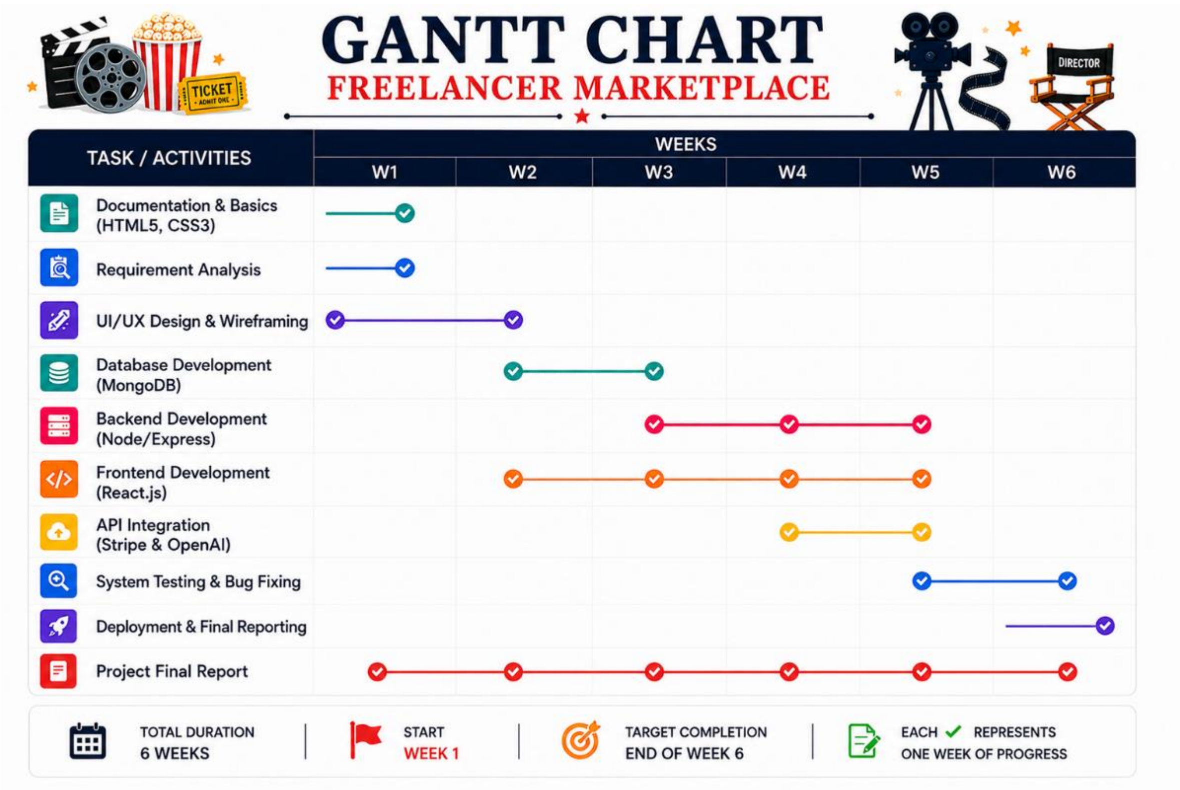
6. Hardware & Software Requirements

- **Hardware Requirements**

Component	Minimum Specification
Processor	Intel i5 / AMD Ryzen 5 or higher.
RAM	8 GB or higher.
Storage	256 GB SSD minimum.
Network	Stable Internet Connection.

- **Software Requirements**

Component	Minimum Specification
Operating System	Windows 10/11, macOS, or Linux.
Code Editor	Visual Studio Code or any modern IDE.
Core Runtime & Tools	Node.js, npm, nodemon, and Postman/thunder client.
Container Engine	Docker Desktop.
Databases	MongoDB.
Third-Party APIs	OpenAI API and Stripe API.
Cloud Hosting	Vercel/Render or AWS EC2.



7. Future Work of the Project

- Freelancer Battle Mode.
- Mobile Application Development.
- Voice and Video Collaboration.

8. References / Bibliography

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